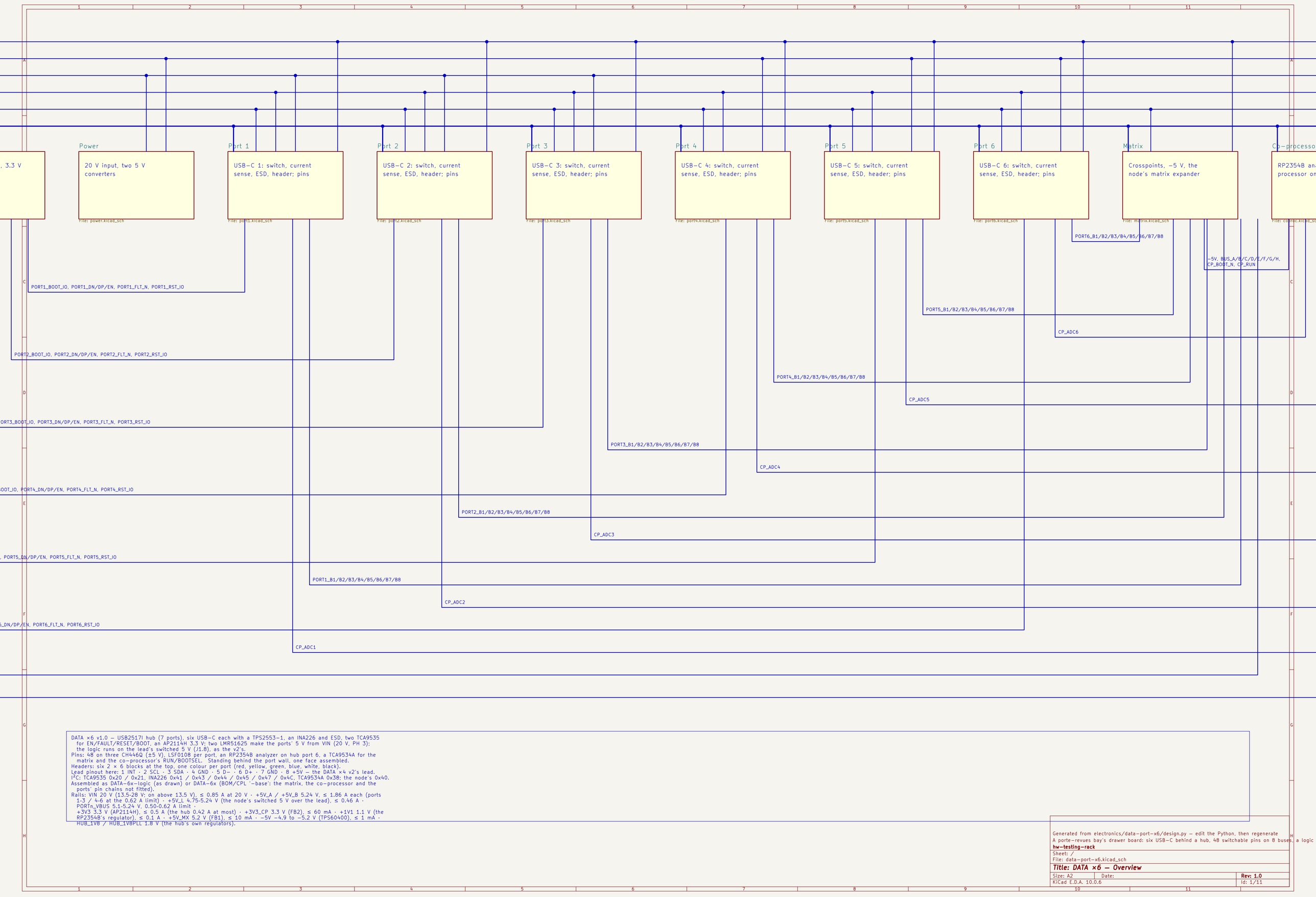


DATA ×6 v1.0 — USB2517I hub (7 ports), six USB-C each with a TPS2553-1, an INA226 and ESD, two TCA9535 for EN/FAULT/RESET/BOOT, an AP2114H 3.3 V; two LMR51625 make the ports' 5 V from VIN (20 V, PH 3); the logic runs on the lead's switched 5 V (J1.8), as the v2's.
Pins: 48 on three CH446Q (±5 V), LSF0108 per port, an RP2354B analyzer on hub port 6, a TCA9534A for the matrix and the co-processor's RUN/BOOTSEL. Standing behind the port wall, one face assembled.
Headers: six 2 × 6 blocks at the top, one colour per port (red, yellow, green, blue, white, black).
Lead pinout here: 1 INT · 2 SCL · 3 SDA · 4 GND · 5 D− · 6 D+ · 7 GND · 8 +5V — the DATA ×4 v2's lead.
I²C: TCA9535 0x20 / 0x21, INA226 0x41 / 0x43 / 0x44 / 0x45 / 0x47 / 0x4C, TCA9534A 0x38; the node's 0x40.
Assembled as DATA-6x-logic (as drawn) or DATA-6x (BOM/CPL '-base': the matrix, the co-processor and the ports' pin chains not fitted).
Rails: VIN 20 V (13.5-28 V; on above 13.5 V), ≤ 0.85 A at 20 V · +5V_A / +5V_B 5.24 V, ≤ 1.86 A each (ports 1-3 / 4-6 at the 0.62 A limit) · +5V_L 4.75-5.24 V (the node's switched 5 V over the lead), ≤ 0.46 A · PORTn_VBUS 5.1-5.24 V, 0.50-0.62 A limit · +3V3 3.3 V (AP2114H), ≤ 0.5 A (the hub 0.42 A at most) · +3V3_CP 3.3 V (FB2), ≤ 60 mA · +1V1 1.1 V (the RP2354B's regulator), ≤ 0.1 A · +5V_MX 5.2 V (FB1), ≤ 10 mA · −5V −4.9 to −5.2 V (TPS60400), ≤ 1 mA · HUB_1V8 / HUB_1V8PLL 1.8 V (the hub's own regulators).

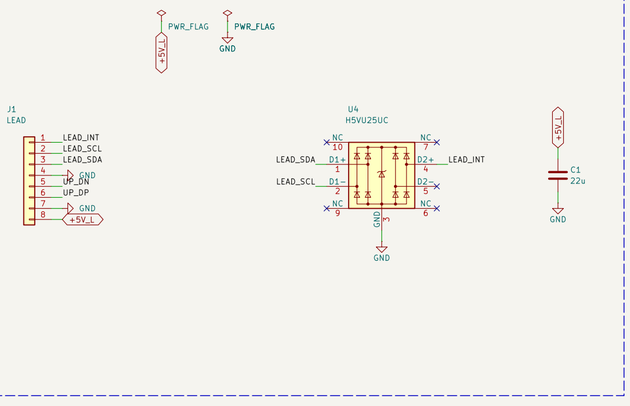
Generated from electronics/data-port-x6/design.py — edit the Python, then regenerate
A porte-revues bay's drawer board: six USB-C behind a hub, 48 switchable pins on 8 buses, a logic
hw-testing-rack

Sheet: /
File: data-port-x6.kicad_sch

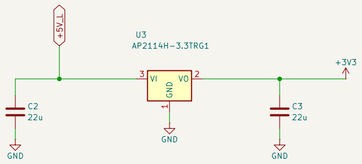
Title: DATA ×6 — Overview

Size: A2	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 1/11

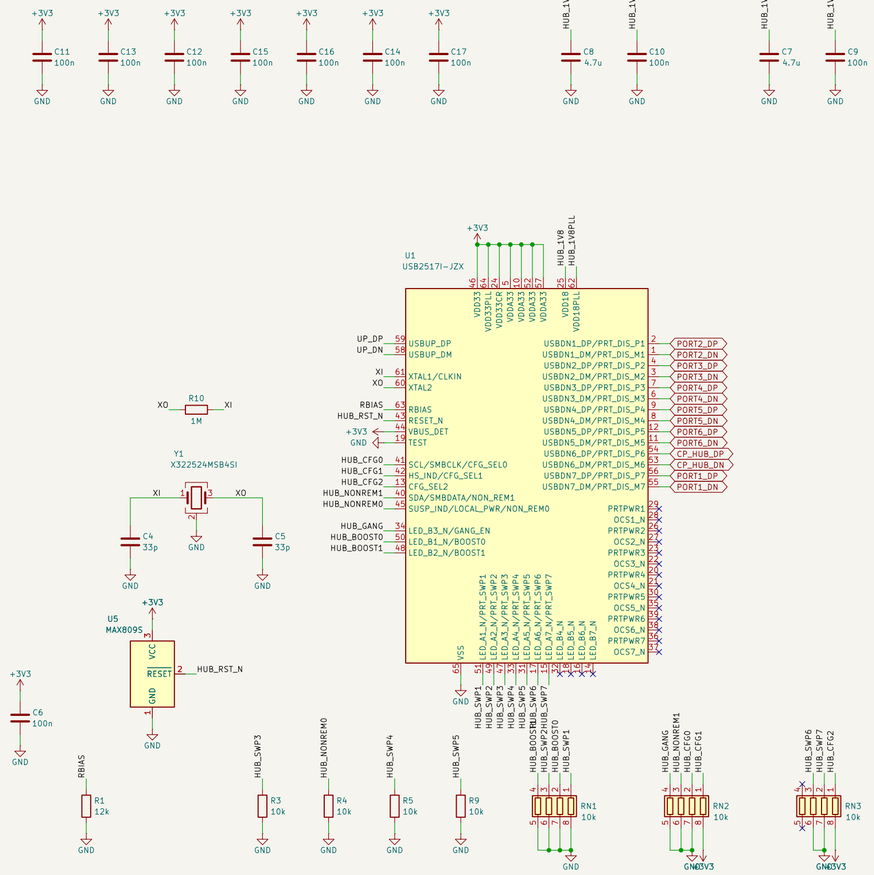
Lead from the node's J7 (PH 8, rear face): I²C and INT through an H5VU25UC; pin 8 the logic's 5 V



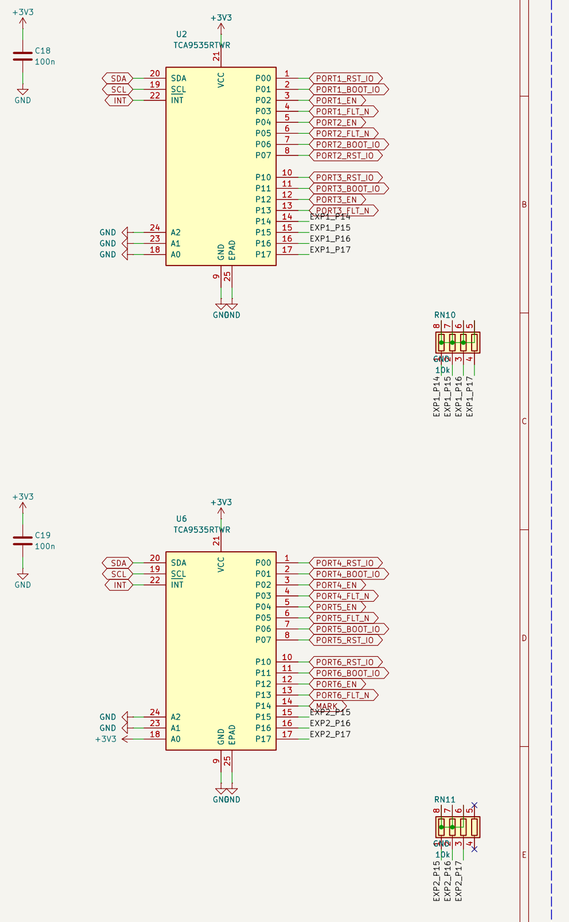
3.3 V for the hub, the expanders, the INA226s and the co-processor (AP2114H, SOT-223 on copper)



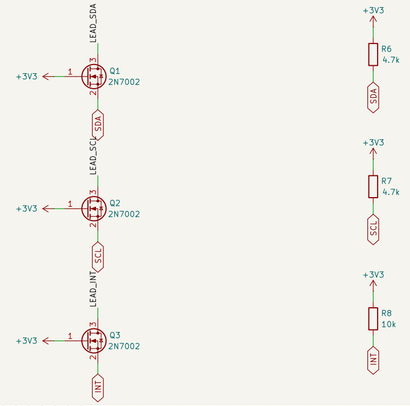
USB2517I hub, CFG_SEL 110 (internal default configuration, straps disabled), 24 MHz crystal



Two TCA9535 expanders, three ports each: EN, FAULT_N, RESET, BOOT; the mark on U6's P14



I²C isolation: open with the board's 3.3 V off (the node's pull-ups can't feed it)



Test points (rear face, bare pads, not in the BOM): +3V3, the hub's reset and 1.8 V, the lead's I²C and INT, GND

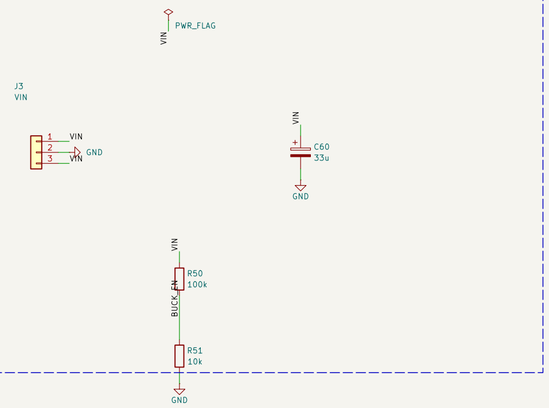


Hub reset: U5 (MAX809S, 2.93 V, ≥ 150 ms, push-pull) on RESET_N (errata DS80000628A, Module 2).
Hub: CFG_SEL[2:0] = 110 (internal default configuration, strap options disabled; DS00001598C Table 5-2).
as the DATA ×4 v2's; all seven downstream ports used.
Hub ports: 1-5 = USB-C 2-6 (busid 1-1.1 … 1-1.5). 6 = the co-processor (1-1.6). 7 = USB-C 1 (1-1.7).
Expanders: TCA9535 U2 0x20 (ports 1-3). U6 0x21 (ports 4-6, the mark on P14). Each port: RESET, BOOT, EN, FAULT_N (port 1 on P00-P03, 2 on P04-P07, 3 on P10-P13). Inputs at power-up: the EN pull-downs keep every port off until the node writes; the spare pins set as outputs, low, by the firmware.
INT (J1.1 → the node's IO4): both expanders' INT and the six INA226 ALERTs, open drain, wired-OR, through Q3.
I²C: TCA9535 0x20 / 0x21, INA226 0x41 / 0x43 / 0x44 / 0x45 / 0x47 / 0x4C, TCA9534A 0x38: the node's INA226 0x40.
J1 pin 8 (the node's switched +5V, ≤ 1.34 A) feeds the logic (+5V_L: U3, +5V_MX): the ports take theirs from VIN (sheet Power). With +5V_L off, the expanders' pins float and the EN pull-downs keep the ports off.
J1 (through-hole) stands on the rear face: soldered by hand (the PCBA places the front face only).

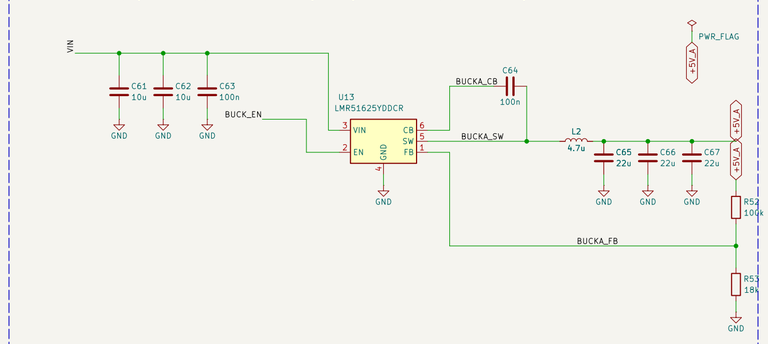
Generated from electronics/data-port-x6/design.py — edit the Python, then regenerate
A porte-revues bay's drawer board: six USB-C behind a hub, 48 switchable pins on 8 buses, a logic

hw-testing-rack		
Sheet: /Hub/		
File: hub.kicad_sch		
Title: DATA ×6 — Lead, hub, expanders, 3.3 V		
Size: A2	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 2/11

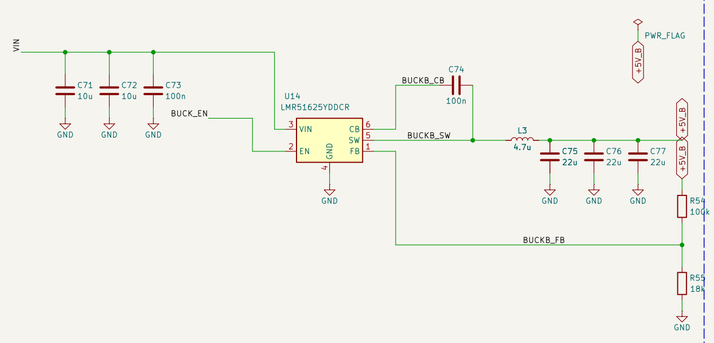
VIN: the rack's 20 V from the node (after its reverse protection), a PH 3 VIN-GND-VIN on the rear face



+5V_A: LMR51625, synchronous, 1.1 MHz, 2.5 A (the bay node's block; TI SLUSF64B)

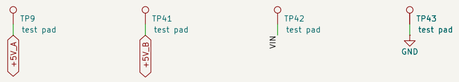


+5V_B: LMR51625, synchronous, 1.1 MHz, 2.5 A (the bay node's block; TI SLUSF64B)



Two LMR51625 as the bay node's (TI SLUSF64B table 8-1: 4.7 µH FXL0630, 2 × 10 µF + 100 nF in, 3 × 22 µF out).
+5V_A: ports 1-3; +5V_B: ports 4-6 (the logic takes +5V_L, J1.8). Never in parallel: the power plane's +5V split left / right. Each 1.5 A at 0.5 A a port, 1.86 A at the TPS2553s' 0.62 A limit (≈ 1 W lost each).
VIN on J3 (PH 3: 1 VIN, 2 GND, 3 VIN): either pre-crimped lead direction fits.
FB 100k/18k: 5.24 V, as the node's. EN 100k/10k: on above 13.5 V, off below 10.8 V.
VIN from the node's VRACK after its reverse-protection P-FET (node rev. B): no second protection here.
≈ 0.95 A at 20 V, 0.7 A at 28 V: a PH contact carries 2 A.

Test points (rear face): +5V_A, +5V_B, VIN, GND



Generated from electronics/data-port-x6/design.py — edit the Python, then regenerate
A porte-revues bay's drawer board: six USB-C behind a hub, 48 switchable pins on 8 buses, a logic

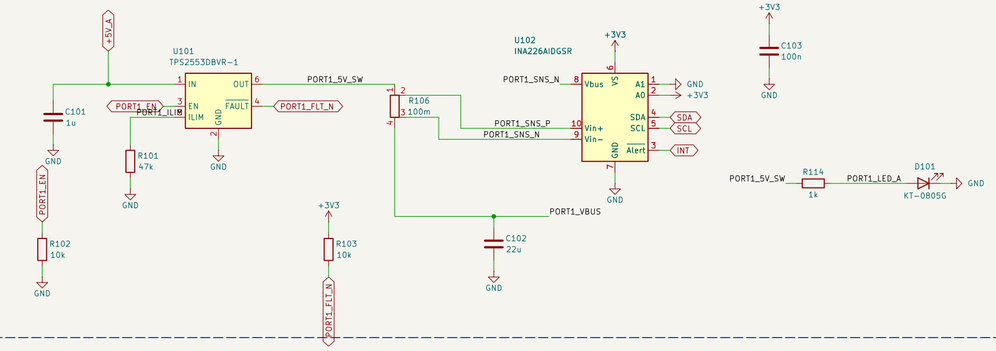
hw-testing-rack

Sheet: /Power/
File: power.kicad_sch

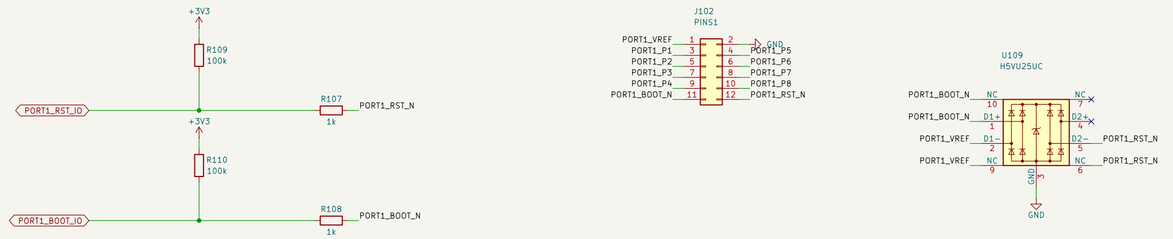
Title: DATA ×6 — 20 V input, two 5 V converters

Size: A2	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 3/11

Port 1 power: +5V_A → TPS2553-1 → PORT1_5V_SW → 100 mΩ shunt → PORT1_VBUS, the INA226 across the shunt

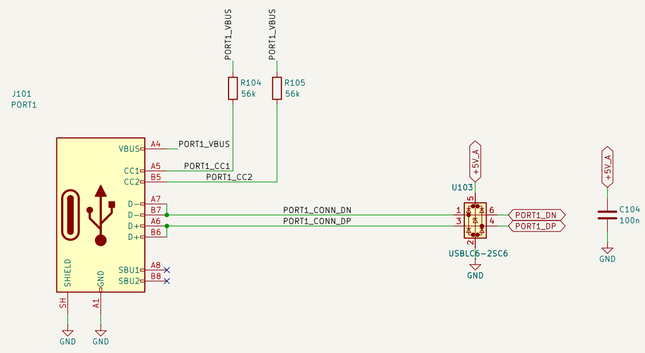


Header 1 (2 × 6 through-hole, red): GND, VREF, P1-P8, RESET_N, BOOT_N; one H5VU25UC on BOOT, VREF and RESET (flow-through: VREF on to the LSF0108 and the divider, RESET/BOOT to the 1k by the expander, their

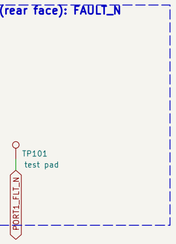


Port 1 = hub port 7, busid 1-1.7; supplied by +5V_A.
INA226 at 0x41; expander U2 (TCA9535, 0x20).
ILIM 47k: 0.50–0.62 A; the ~1 latches off after 7.5 ms in limit, FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → LSF0108 ch5 6 7 8 4 3 2 1 (LANE) → crosspoint X lines (XMAP, the DATA ×4 v2's;
a third chip for ports 5-6). An unwired VREF stays ≈ 0.4 V, under the LSF0108's 0.65 V: the
firmware keeps the port's crosspoint switches open while its VREF reads under 0.65 V.
ESD (flow-through): P1-P4 U105, P5-P8 U106, BOOT_N / VREF / RESET_N U109.
VREF: the board's I/O supply, 0.95–3.6 V; a 5 V board with TTL inputs: its own 3.3 V pin.
Source only (Rp from the switched VBUS); a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≈ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
Header: red; left column VREF 1 2 3 4 BOOT, right GND 5 6 7 8 RESET.

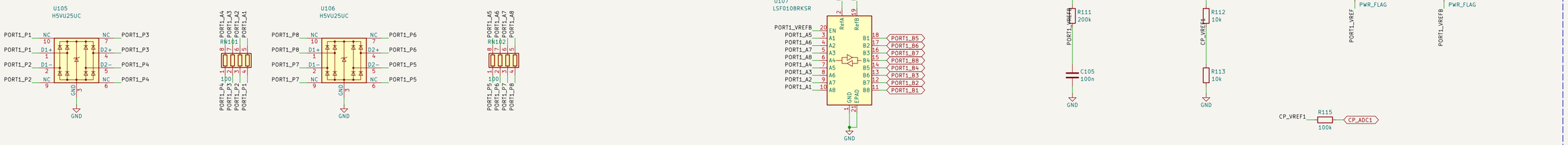
USB-C 1 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



Test point (rear face): FAULT_N



Pins 1: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U7's X lines

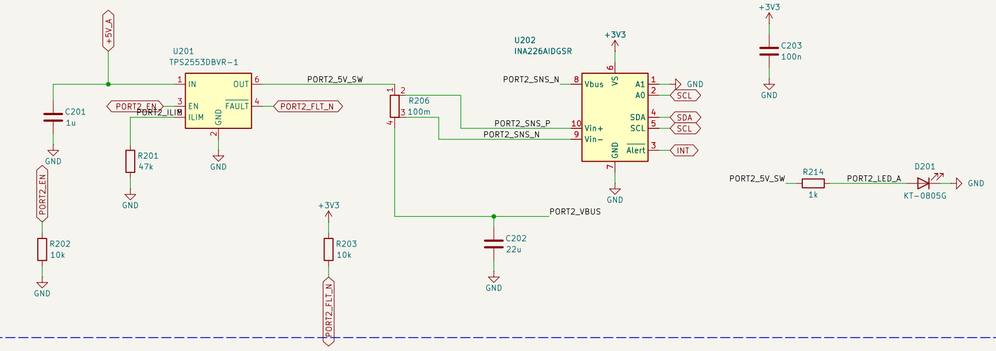


Generated from electronics/data-port-x6/design.py — edit the Python, then regenerate
A porte-revues bay's drawer board: six USB-C behind a hub, 48 switchable pins on 8 buses, a logic
hw-testing-rack
Sheet: /Port 1/
File: port1.kicad_sch

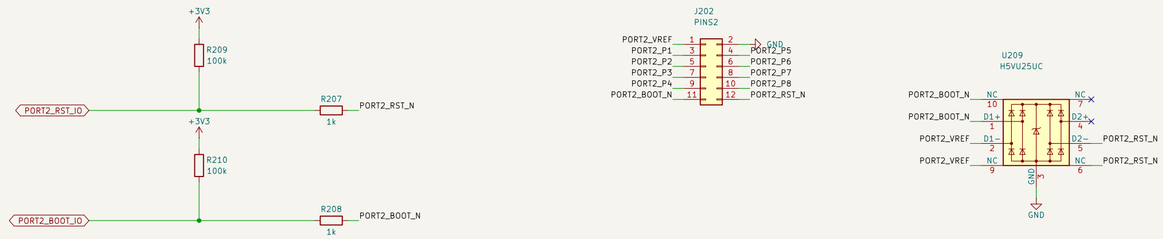
Title: DATA ×6 — USB-C 1: switch, current sense, ESD, header; pins

Size: A2	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 4/11

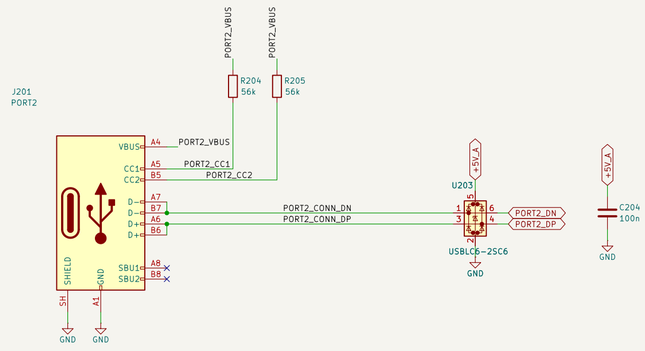
Port 2 power: +5V_A → TPS2553-1 → PORT2_5V_SW → 100 mΩ shunt → PORT2_VBUS, the INA226 across the shunt



Header 2 (2 × 6 through-hole, yellow): GND, VREF, P1-P8, RESET_N, BOOT_N; one H5VU25UC on BOOT, VREF and RESET (flow-through: VREF on to the LSF0108 and the divider, RESET/BOOT to the 1k by the expander, th

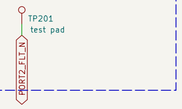


USB-C 2 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)

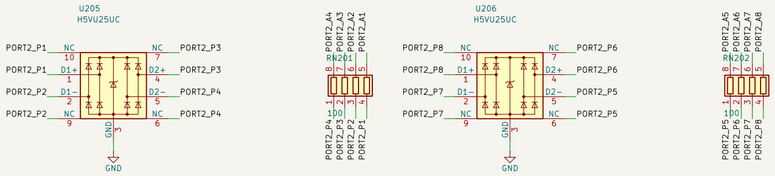


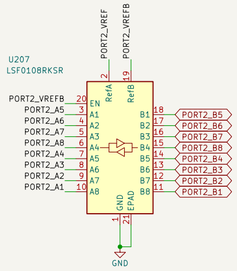
Port 2 = hub port 1, busid 1-1.1: supplied by +5V_A.
INA226 at 0x43; expander U2 (TCA9535, 0x20).
ILIM 47k: 0.50-0.62 A; the ~1 latches off after 7.5 ms in limit, FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → LSF0108 ch5 6 7 8 4 3 2 1 (LANE) → crosspoint X lines (XMAP, the DATA ×4 v2's;
a third chip for ports 5-6). An unwired VREF stays ≈ 0.4 V, under the LSF0108's 0.65 V: the
firmware keeps the port's crosspoint switches open while its VREF reads under 0.65 V.
ESD (flow-through): P1-P4 U205, P5-P8 U206, BOOT_N / VREF / RESET_N U209.
VREF: the board's I/O supply, 0.95-3.6 V; a 5 V board with TTL inputs: its own 3.3 V pin.
Source only (Rp from the switched VBUS); a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≥ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
Header: yellow: left column VREF 1 2 3 4 BOOT, right GND 5 6 7 8 RESET.

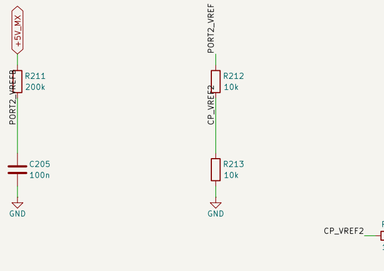
Test point (rear face): FAULT_N



Pins 2: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U7's X lines





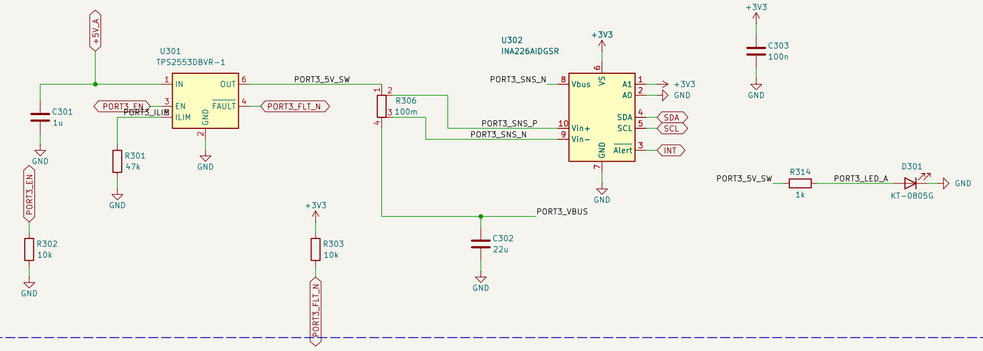


Generated from electronics/data-port-x6/design.py — edit the Python, then regenerate
A porte-revues bay's drawer board: six USB-C behind a hub, 48 switchable pins on 8 buses, a logic
hw-testing-rack

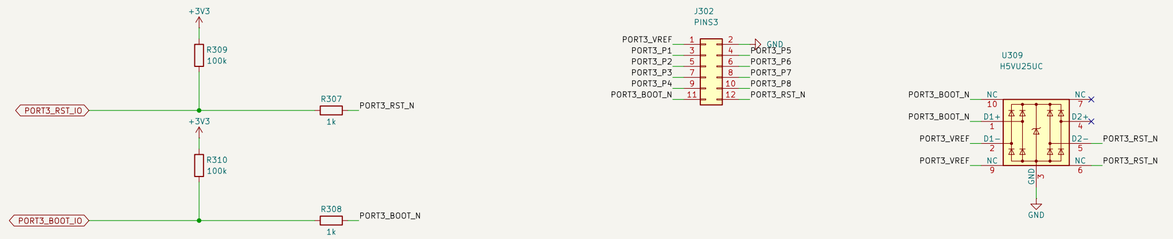
Sheet: /Port 2/	
File: port2.kicad_sch	
Title: DATA ×6 — USB-C 2: switch, current sense, ESD, header; pins	
Size: A2	Date:
KiCad E.D.A. 10.0.6	Id: 5/11

Rev: 1.0

Port 3 power: +5V_A → TPS2553-1 → PORT3_5V_SW → 100 mΩ shunt → PORT3_VBUS, the INA226 across the shunt

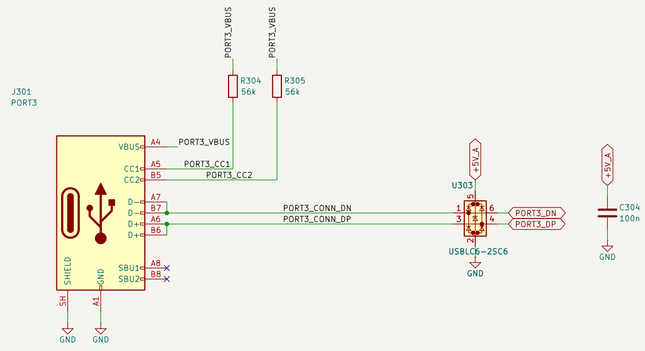


Header 3 (2 × 6 through-hole, green): GND, VREF, P1-P8, RESET_N, BOOT_N; one H5VU25UC on BOOT, VREF and RESET (flow-through: VREF on to the LSF0108 and the divider, RESET/BOOT to the 1k by the expander, the

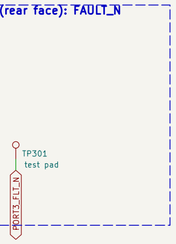


Port 3 = hub port 2, busid 1-1.2; supplied by +5V_A.
INA226 at 0x44; expander U2 (TCA9535, 0x20).
ILIM 47k: 0.50-0.62 A; the ~1 latches off after 7.5 ms in limit, FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → LSF0108 ch5 6 7 8 4 3 2 1 (LANE) → crosspoint X lines (XMAP, the DATA ×4 v2's;
a third chip for ports 5-6). An unwired VREF stays ≈ 0.4 V, under the LSF0108's 0.65 V: the
firmware keeps the port's crosspoint switches open while its VREF reads under 0.65 V.
ESD (flow-through): P1-P4 U305, P5-P8 U306, BOOT_N / VREF / RESET_N U309.
VREF: the board's I/O supply, 0.95-3.6 V; a 5 V board with TTL inputs: its own 3.3 V pin.
Source only (Rp from the switched VBUS); a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≈ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
Header: green; left column VREF 1 2 3 4 BOOT, right GND 5 6 7 8 RESET.

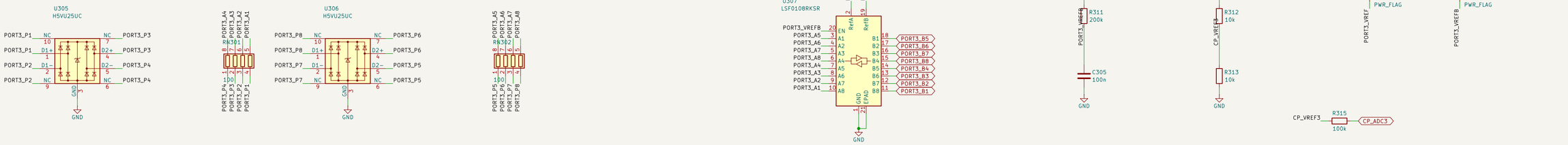
USB-C 3 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



Test point (rear face): FAULT_N



Pins 3: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U8's X lines



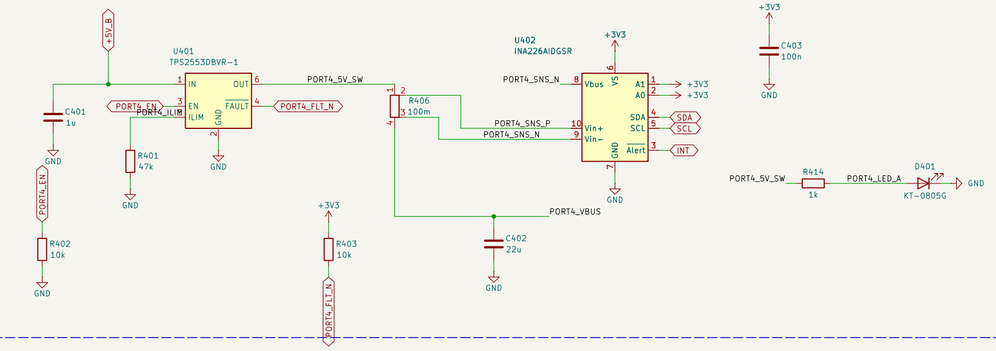
Generated from electronics/data-port-x6/design.py — edit the Python, then regenerate
A porte-revues bay's drawer board: six USB-C behind a hub, 48 switchable pins on 8 buses, a logic
hw-testing-rack

Sheet: /Port 3/
File: port3.kicad_sch

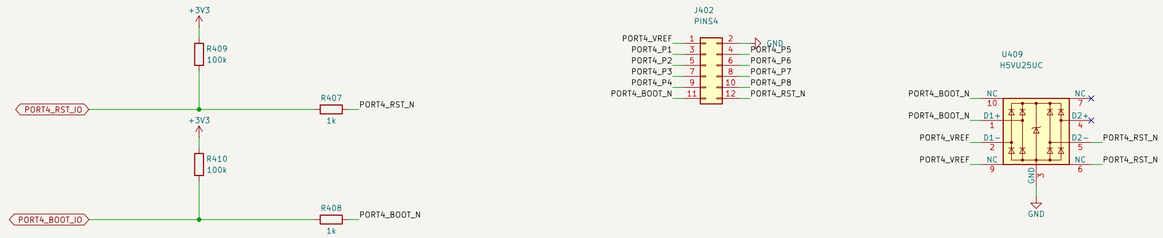
Title: DATA ×6 — USB-C 3: switch, current sense, ESD, header; pins

Size: A2	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 6/11

Port 4 power: +5V_B → TPS2553-1 → PORT4_5V_SW → 100 mΩ shunt → PORT4_VBUS, the INA226 across the shunt

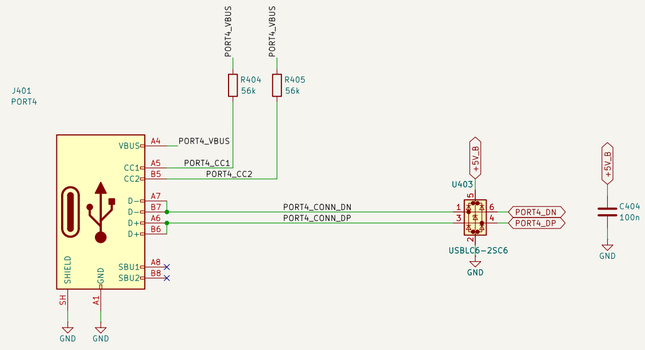


Header 4 (2 × 6 through-hole, blue): GND, VREF, P1-P8, RESET_N, BOOT_N; one H5VU25UC on BOOT, VREF and RESET (flow-through: VREF on to the LSF0108 and the divider, RESET/BOOT to the 1k by the expander, the

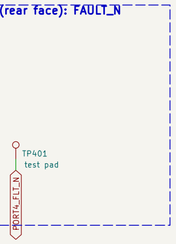


Port 4 = hub port 3, busid 1-1.3; supplied by +5V_B.
INA226 at 0x45; expander U6 (TCA9535, 0x21).
ILIM 47k: 0.50-0.62 A; the ~1 latches off after 7.5 ms in limit, FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → LSF0108 ch5 6 7 8 4 3 2 1 (LANE) → crosspoint X lines (XMAP, the DATA ×4 v2's;
a third chip for ports 5-6). An unwired VREF stays ≈ 0.4 V, under the LSF0108's 0.65 V: the
firmware keeps the port's crosspoint switches open while its VREF reads under 0.65 V.
ESD (flow-through): P1-P4 U405, P5-P8 U406, BOOT_N / VREF / RESET_N U409.
VREF: the board's I/O supply, 0.95-3.6 V; a 5 V board with TTL inputs: its own 3.3 V pin.
Source only (Rp from the switched VBUS); a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≥ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
Header: blue: left column VREF 1 2 3 4 BOOT, right GND 5 6 7 8 RESET.

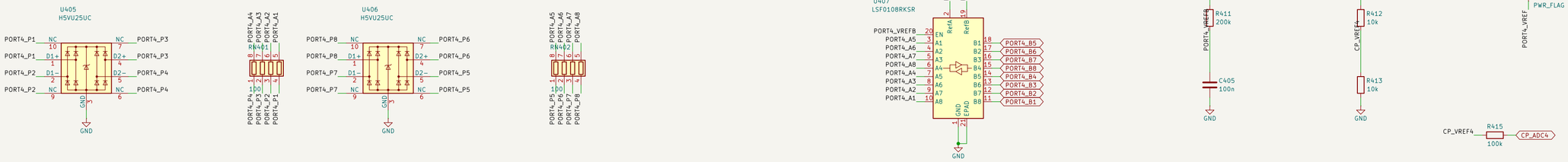
USB-C 4 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



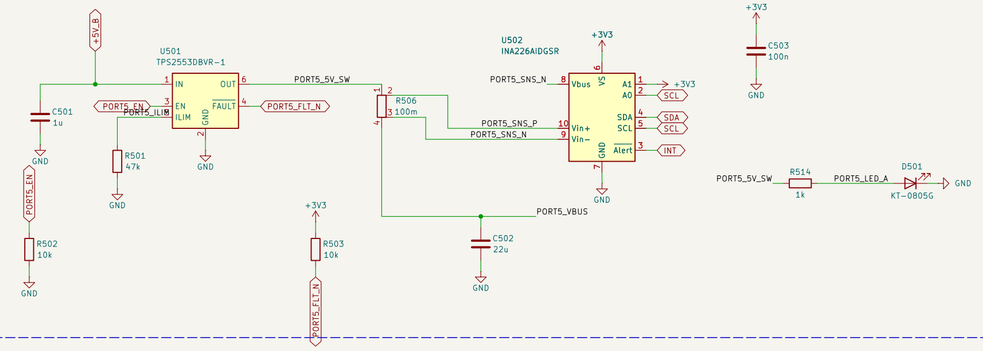
Test point (rear face): FAULT_N



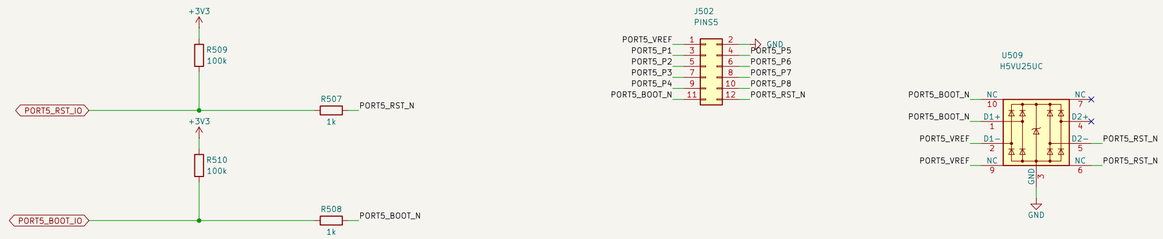
Pins 4: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U8's X lines



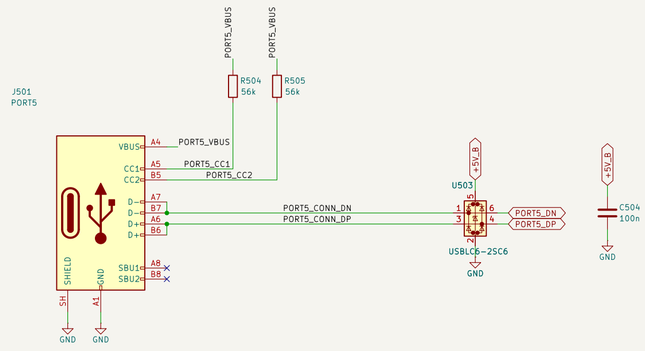
Port 5 power: +5V_B → TPS2553-1 → PORT5_5V_SW → 100 mΩ shunt → PORT5_VBUS, the INA226 across the shunt



Header 5 (2 × 6 through-hole, white): GND, VREF, P1-P8, RESET_N, BOOT_N; one H5VU25UC on BOOT, VREF and RESET (flow-through: VREF on to the LSF0108 and the divider, RESET/BOOT to the 1k by the expander, the

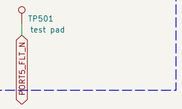


USB-C 5 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)

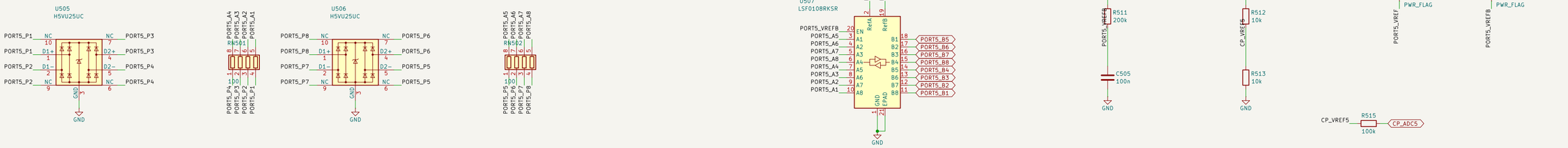


Port 5 = hub port 4, busid 1-1.4; supplied by +5V_B.
INA226 at 0x47: expander U6 (TCA9535, 0x21).
ILIM 47k: 0.50-0.62 A; the ~1 latches off after 7.5 ms in limit, FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → LSF0108 ch5 6 7 8 4 3 2 1 (LANE) → crosspoint X lines (XMAP, the DATA ×4 v2's;
a third chip for ports 5-6). An unwired VREF stays ≈ 0.4 V, under the LSF0108's 0.65 V: the
firmware keeps the port's crosspoint switches open while its VREF reads under 0.65 V.
ESD (flow-through): P1-P4 U505, P5-P8 U506, BOOT_N / VREF / RESET_N U509.
VREF: the board's I/O supply, 0.95-3.6 V; a 5 V board with TTL inputs: its own 3.3 V pin.
Source only (Rp from the switched VBUS); a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≥ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
Header: white; left column VREF 1 2 3 4 BOOT, right GND 5 6 7 8 RESET.

Test point (rear face): FAULT_N

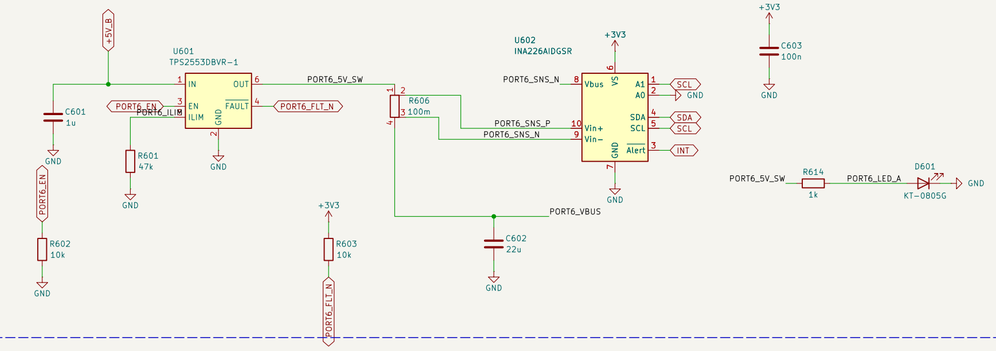


Pins 5: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U12's X lines

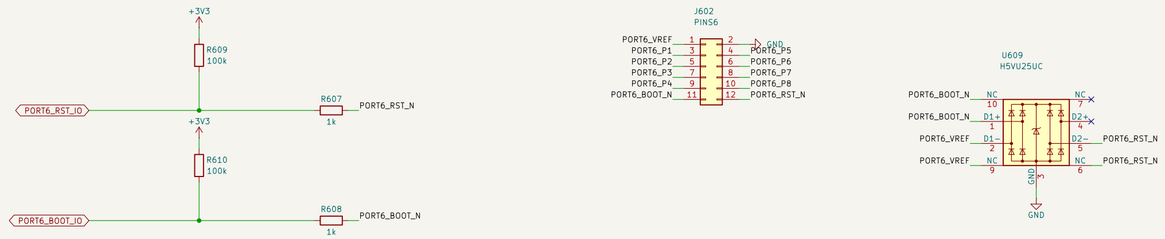


Generated from electronics/data-port-x6/design.py — edit the Python, then regenerate
A porte-revues bay's drawer board: six USB-C behind a hub, 48 switchable pins on 8 buses, a logic
hw-testing-rack
Sheet: /Port 5/
File: port5.kicad_sch
Title: DATA ×6 — USB-C 5: switch, current sense, ESD, header; pins
Size: A2 | Date: | **Rev: 1.0**
KiCad E.D.A. 10.0.6 | Id: 8/11

Port 6 power: +5V_B → TPS2553-1 → PORT6_5V_SW → 100 mΩ shunt → PORT6_VBUS, the INA226 across the shunt

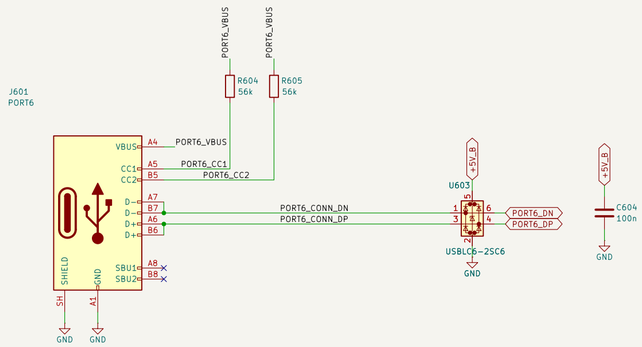


Header 6 (2 × 6 through-hole, black): GND, VREF, P1-P8, RESET_N, BOOT_N; one H5VU25UC on BOOT, VREF and RESET (flow-through: VREF on to the LSF0108 and the divider, RESET/BOOT to the 1k by the expander, the

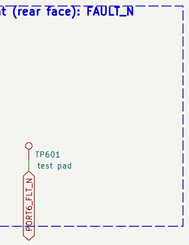


Port 6 = hub port 5, busid 1-1.5: supplied by +5V_B.
INA226 at 0x4c: expander U6 (TCA9535, 0x21).
ILIM 47k: 0.50-0.62 A; the ~1 latches off after 7.5 ms in limit, FAULT_N flags it.
INA226 Vbus on the shunt's load-side sense pad (= VBUS).
EN pulled down: the port stays off until the node switches it on.
Pins: P1-P8 → LSF0108 ch5 6 7 8 4 3 2 1 (LANE) → crosspoint X lines (XMAP, the DATA ×4 v2's;
a third chip for ports 5-6). An unwired VREF stays ≈ 0.4 V, under the LSF0108's 0.65 V: the
firmware keeps the port's crosspoint switches open while its VREF reads under 0.65 V.
ESD (flow-through): P1-P4 U605, P5-P8 U606, BOOT_N / VREF / RESET_N U609.
VREF: the board's I/O supply, 0.95-3.6 V; a 5 V board with TTL inputs: its own 3.3 V pin.
Source only (Rp from the switched VBUS); a back-feeding source trips the TPS2553 (latched).
The DUT's RESET/BOOT pull-ups ≥ 4.7k: against 1k, the 1k in series stops at ≈ 1.65 V.
Header: black; left column VREF 1 2 3 4 BOOT, right GND 5 6 7 8 RESET.

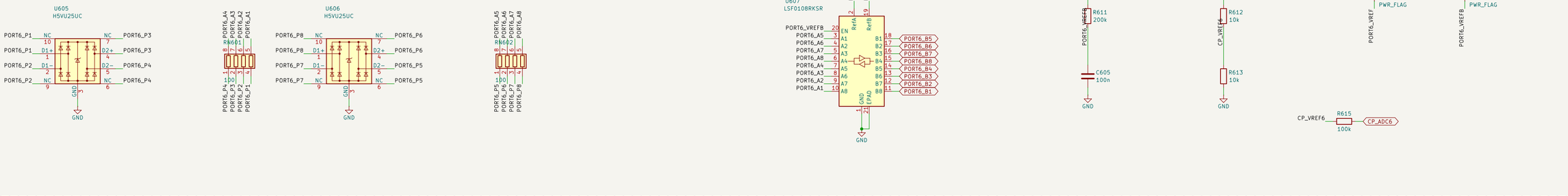
USB-C 6 (source): Rp 56k on each CC; USBLC6 in flow-through (connector side | hub side)



Test point (rear face): FAULT_N



Pins 6: ESD in flow-through, 100 Ω, LSF0108 (A = the DUT's VREF, B = 3.3 V) to crosspoint U12's X lines



Generated from electronics/data-port-x6/design.py — edit the Python, then regenerate
A porte-revues bay's drawer board: six USB-C behind a hub, 48 switchable pins on 8 buses, a logic

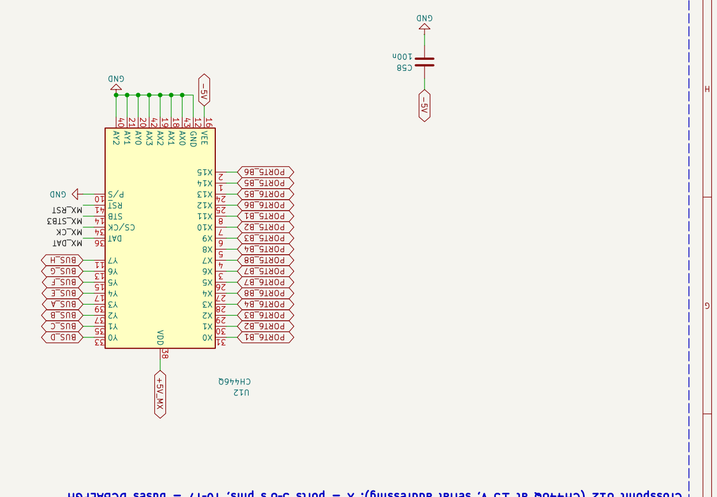
hw-testing-rack

Sheet: /Port 6/
File: port6.kicad_sch

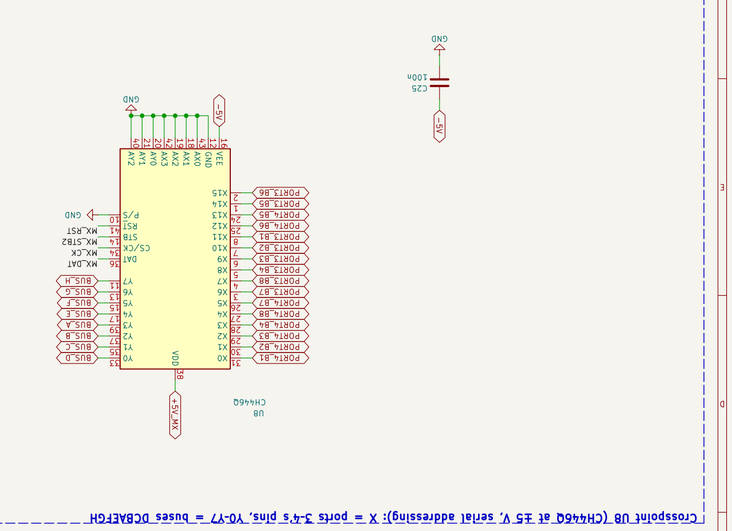
Title: DATA ×6 — USB-C 6: switch, current sense, ESD, header; pins

Size: A2	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 9/11

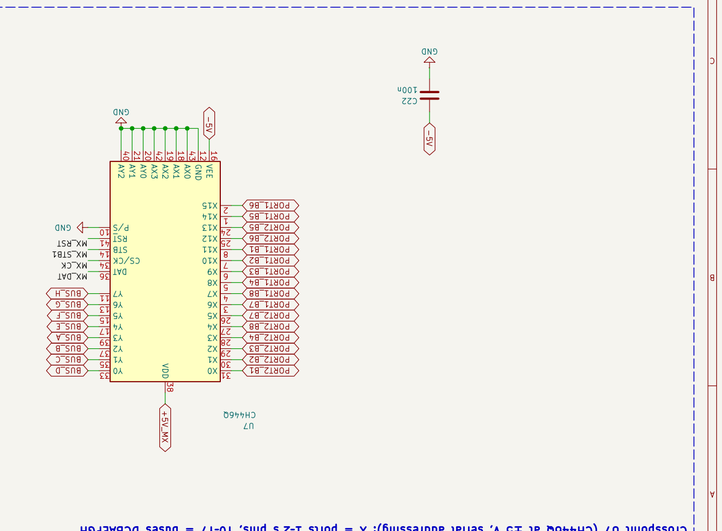
Crosspoint U12 (CH446Q at I²C 45 V, serial addressing): X = ports 5-6's pins, Y0-Y7 = buses DCBAEFGH



Crosspoint U8 (CH446Q at I²C 45 V, serial addressing): X = ports 3-4's pins, Y0-Y7 = buses DCBAEFGH

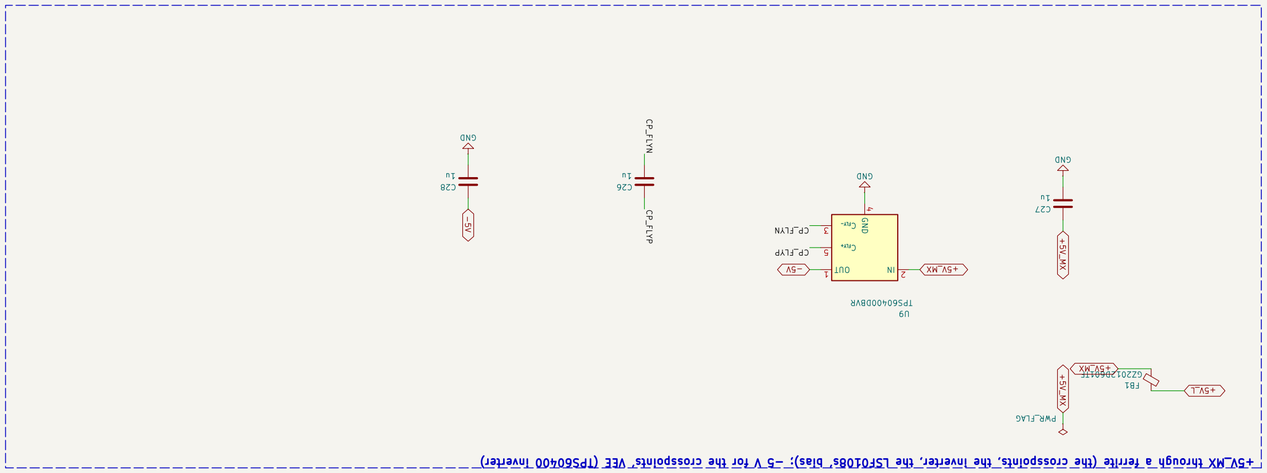


Crosspoint U7 (CH446Q at I²C 45 V, serial addressing): X = ports 1-2's pins, Y0-Y7 = buses DCBAEFGH

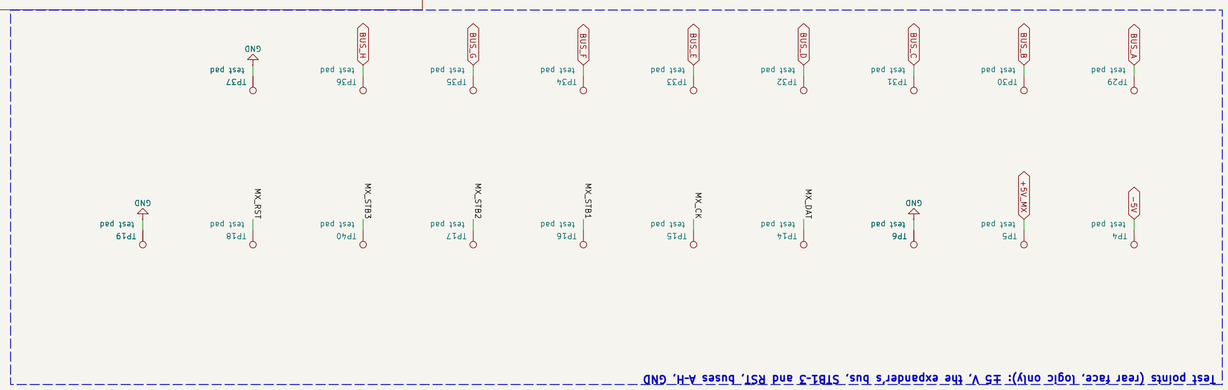


The node's matrix expander (TCA9534A, 0x38): the crosspoints and the co-processor's RUN / BOOTSEL

+5V_MX through a ferrite (the crosspoints, the inverter, the LSF0108' bias): -5 V for the crosspoints' VEE (TPS60400 inverter)



Test points (rear face, logic only): ±5 V, the expander's bus, STB1-3 and RST, buses A-H, GND



Generated from electronics/data-port-x6/design.py — edit the Python, then regenerate
A porte-revues bay's drawer board: six USB-C behind a hub, 48 switchable pins on 8 buses, a logic

hw-testing-rack

Sheet: /Matrix/
File: matrix.kicad_sch

Title: DATA ×6 — Crosspoints, −5 V, the node's matrix expander

Size: A2	Date:
KiCad E.D.A. 10.0.6	Rev: 1.0
	Id: 10/11

SWD header (J2, 1 × 5, by hand): Debug Probe 1-3

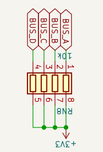


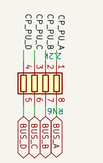
Test points (rear face, logic only): SWD, RUN, BOOTSEL, rails, GND

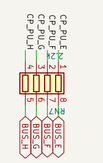


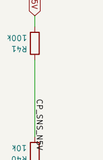
GPIO2-9 = buses A-H (gusmanb CH1-8, B-bit mode); GPIO0 → GPIO1 bridged (its trigger path).
GPIO14 = the node's mark (TCA9555 U6 P14); GPIO25 = LED; GPIO19...16, 13...10 = pull-up enables A...H (2.2k):
GPIO40, 42, 52, 41, 44, 45 = VREF 1-6 / 7 (through 100Ω); GPIO46 = +5V_MX × 47/147;
GPIO47 = 3.3 V − (3.3 + 5) × 10/110 from −5 V (2.95 V nominal). The ADC's eight inputs all used.
RUN pulled up. BOOTSEL through 1k; both held by the node (TCA9554A). open drain.
USB on the hub's port 6 (Full Speed, 22 Ω).
Core regulator: Raspberry Pi's minimal RA block turned 180°: L1's dot (pad 1) on +1V1, not VREG_LX.
C47 (DVDD pin 32, furthest from the regulator) 4.7 µF; no copper under LX / L1 (In1. In2. B.Cu).

Bus pull-ups: 10k to 3.3 V always (the LSFs' B side), 2.2k switched by the co-processor (GPIO10-13, 16-19); the ±5 V rails into the ADC



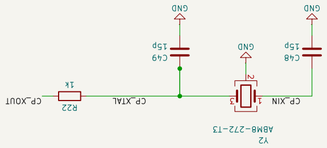






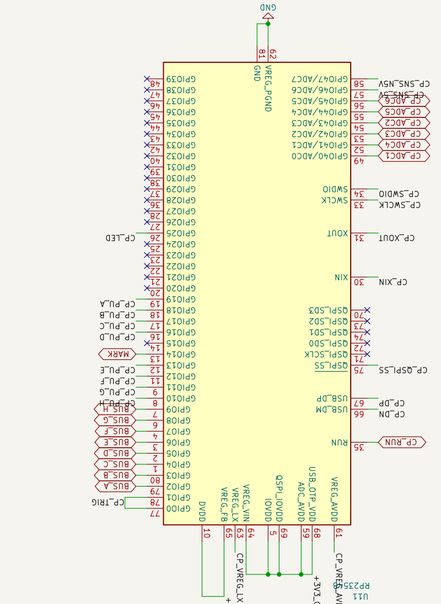
12 MHz crystal (ABM8-272-T3, 15 pF, 1k on XOUT), USB on hub port 6 (22 Ω), RUN, BOOTSEL, LED, SWD

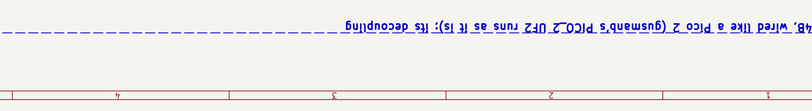




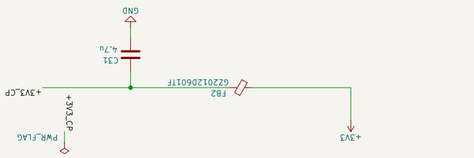


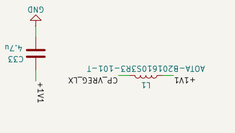
RP2354B, wired like a Pico 2 (gusmanb's PICO_2_UF2 runs as it is); its decoupling





It's 3.3 V from +3V3 through a ferrite, the core regulator as Raspberry Pi lays it out (3.3 µH, 3 × 4.7 µF, 33 Ω)





Generated from electronics/data-port-x6/design.py — edit the Python, then regenerate
A porte-revues bay's drawer board: six USB-C behind a hub, 48 switchable pins on 8 buses, a logic
hw-testing-rack

Sheet: /Co-processor/
File: coprocx.kicad_sch

Title: DATA ×6 — RP2354B analyzer / co-processor on hub port 6

Size: A2	Date:
KiCad E.D.A. 10.0.6	Rev: 1.0
	Id: 11/11